

Name: _____

Date: _____

PLACE VALUE AND ORDERING INTEGERS, (READING AND WRITING LARGE INTEGERS)

LO: I can read, write and compare large integers up to millions using place value.

Large integers

Place value tells us the value of each digit based on its position. Large numbers use thousands, millions, and billions columns.

Commas: We use commas (or spaces) to separate groups of three digits from the right, making large numbers easier to read.

Quick examples

●	3,500,000 = three million five hundred thousand	reading correctly
●	45,000 has 4 in the ten-thousands column	place value
●	1,000,000 = one million	six zeros
●	3,500,000 = thirty-five thousand	wrong place value
●	1,000,000 has 5 zeros	it has 6 zeros

Key idea

Use place value columns: ones, tens, hundreds, thousands, ten-thousands, hundred-thousands, millions.

2,456,789

2 millions, 456 thousands, 789 ones

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - reading a large number

Write 4,250,000 in words.

4 is in the millions column

250 is in the thousands

Four million two hundred and fifty thousand

= Four million two hundred and fifty thousand

Read from left to right using place value.

Example 2 - writing in figures

Write seven hundred and thirty-two thousand in figures.

Seven hundred and thirty-two thousand = 732 thousand
= 732,000

Add three zeros for thousands

= 732,000

Match each word to its place value position.

Example 3 - identifying digit value

What is the value of the 6 in 3,604,500?

Position: millions column has 3,
hundred-thousands has 6
The 6 is in the hundred-thousands column

Value = 600,000

= 600,000

The position of the digit determines its value.

Example 4 - comparing large numbers

Which is larger: 2,450,000 or 2,540,000?

Both have 2 million
Compare hundred-thousands: 4 vs 5

$5 > 4$, so 2,540,000 is larger

2,540,000 is larger

Compare digits from left to right until they differ.

YOUR TURN – PRACTICE (deliberate practice)

1. Writing numbers in words

Write each number in words.

- a) 35,000
- b) 420,000
- c) 1,500,000
- d) 7,080,000

2. Writing numbers in figures

Write each number in figures.

- a) Sixty thousand
- b) Three hundred and forty-five thousand
- c) Two million
- d) Nine million, two hundred thousand

3. Place value of digits

State the value of the underlined digit (shown in brackets).

- a) The 5 in 2,500,000
- b) The 3 in 4,350,000
- c) The 8 in 8,040,000
- d) The 7 in 1,070,500

4. Ordering large numbers

Put in order from smallest to largest.

- a) 45000, 54000, 40500, 50400
- b) 1200000, 210000, 2100000, 120000
- c) 3050000, 3500000, 305000, 350000
- d) 999999, 1000000, 99999, 1000001

5. Mixed large number problems

Solve each problem.

- a) Write 2.5 million in figures
- b) How many thousands in 3,400,000?
- c) Round 4,678,000 to the nearest million
- d) What is 100,000 more than 2,950,000?
- e) Write 850,000 in words

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) 3,500,000 is three million five hundred thousand ☐

Correct answer: _____

Reason: _____

b) 1,000,000 has 5 zeros ☐

Correct answer: _____

Reason: _____

c) The 4 in 4,200,000 is worth 4 million ☐

Correct answer: _____

Reason: _____

d) 450,000 is bigger than 1,000,000 ☐

Correct answer: _____

Reason: _____

e) We read large numbers from left to right ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) The population of a city is **2,847,300**. Round this to the nearest: (i) thousand, (ii) ten thousand, (iii) hundred thousand, (iv) million.

Expression: _____

Simplified: _____

b) A country has a national debt of **£2,400,000,000** (2.4 billion). If the country pays off **£150,000,000** per year, how many years will it take to pay off the debt? Write your answer using full figures.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

